

DARLINGTON COUNTY AP, SC, USA

WMO: 723116

Lat: 34.450N

Lon: 79.883W

Elev: 192

StdP: 14.59

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 03742

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.8	26.5	7.0	7.9	29.0	10.3	9.4	32.9	24.6	55.2	21.6	54.9	4.1	350	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	96.0	76.3	93.1	75.5	91.1	75.3	80.2	89.4	78.9	88.3	77.9	87.3	8.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.8	145.6	84.7	76.2	137.8	83.1	75.1	132.7	82.0	43.8	89.7	42.5	88.4	41.2	87.4	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.8	17.2	15.4	DB	14.4	99.1	3.9	3.1	11.6	101.4	9.3	103.2	7.1	104.9	4.3	107.1
			WB	13.5	81.5	3.8	2.1	10.8	83.0	8.6	84.2	6.5	85.3	3.8	86.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.4	44.2	47.3	54.4	63.4	72.1	79.4	80.5	79.6	74.6	63.9	53.2	47.1
	DBStd	15.13	10.39	8.91	9.00	7.83	6.20	4.10	3.61	3.99	5.31	7.89	8.28	9.18
	HDD50	684	245	146	59	4	0	0	0	0	0	3	58	169
	HDD65	2670	648	498	344	121	16	0	0	0	6	117	361	559
	CDD50	5572	63	71	195	407	685	881	945	919	738	434	155	79
	CDD65	2082	1	3	15	74	236	431	480	454	293	83	8	4
	CDH74	19019	6	21	152	757	2141	4166	4653	4143	2271	633	65	11
	CDH80	8014	0	1	20	212	816	1921	2125	1889	869	156	5	0
Wind	WSAvg	6.6	6.9	7.4	7.9	8.3	7.5	6.5	5.9	5.1	5.7	5.7	5.7	6.1
Precipitation	PrecAvg	45.8	3.3	3.1	3.5	2.9	3.1	4.7	5.2	5.7	4.6	3.5	2.9	3.2
	PrecMax	58.3	7.1	7.4	6.8	5.8	6.2	9.1	8.3	13.0	9.3	11.1	6.9	8.3
	PrecMin	30.5	0.7	1.2	0.3	1.2	1.5	1.0	1.4	2.1	0.9	0.0	0.4	0.8
	PrecStd	6.1	1.9	1.5	1.6	1.3	1.1	2.1	2.0	2.9	2.6	2.4	1.7	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.4	77.5	82.1	90.0	94.9	98.9	98.7	98.9	94.1	88.4	80.9	75.1
		MCWB	63.4	65.0	64.8	67.4	72.7	76.1	77.9	77.4	73.9	74.0	67.5	66.6
	2%	DB	69.9	72.0	78.9	85.6	90.9	94.9	94.7	94.9	90.9	84.0	75.2	71.8
		MCWB	61.5	61.1	63.2	66.9	71.3	75.2	77.3	77.0	74.2	71.5	63.3	64.7
	5%	DB	64.7	67.2	74.5	81.9	88.2	92.5	92.0	91.5	88.5	81.1	72.2	66.7
		MCWB	58.9	58.6	60.6	64.7	70.5	74.6	77.2	76.3	73.7	68.1	62.4	61.8
	10%	DB	61.2	63.4	70.4	78.9	84.2	90.1	90.3	89.9	85.2	77.2	68.2	63.4
		MCWB	55.1	54.8	58.7	63.6	69.1	74.2	76.7	76.2	73.4	68.3	60.6	58.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.5	67.2	68.0	72.3	76.3	79.5	82.1	82.2	78.5	77.0	70.6	68.7
		MCDB	70.8	74.3	77.6	80.7	86.2	91.2	90.6	91.4	86.2	83.5	75.4	73.3
	2%	WB	63.3	64.2	65.7	69.7	74.3	78.1	80.5	80.3	77.1	74.6	67.3	65.9
		MCDB	68.4	69.3	75.0	79.9	83.8	89.7	89.8	88.8	85.8	80.4	72.4	70.0
	5%	WB	60.0	60.4	63.4	67.7	72.8	77.1	79.2	78.8	75.9	72.3	64.9	63.0
		MCDB	64.1	65.4	70.3	78.1	82.6	88.0	88.8	87.5	84.1	77.6	69.3	65.8
	10%	WB	56.4	56.8	61.4	65.7	71.3	75.9	78.0	77.7	74.7	69.6	62.1	59.1
		MCDB	60.3	62.2	68.3	75.2	81.0	85.4	87.2	86.7	82.0	75.1	67.6	63.4
Mean Daily Temperature Range	5% DB	MDBR	20.1	21.2	23.0	23.5	21.6	19.3	18.7	19.0	19.4	21.5	23.1	20.2
		MCDBR	23.2	25.4	26.7	27.0	24.8	22.5	21.4	22.2	23.2	23.7	25.9	22.6
		MCWBR	16.2	16.5	14.1	11.5	8.9	7.4	7.8	8.2	9.3	11.9	15.8	16.6
	5% WB	MCDBR	19.9	22.9	22.3	22.3	20.1	19.5	19.0	19.2	18.5	17.6	20.4	19.7
		MCWBR	15.4	17.4	13.9	11.2	8.2	7.6	8.1	8.4	8.1	10.0	15.0	16.6
Clear-Sky Solar Irradiance	taub	0.317	0.326	0.361	0.406	0.467	0.493	0.544	0.515	0.432	0.391	0.339	0.313	
	taud	2.500	2.469	2.402	2.265	2.147	2.117	1.998	2.069	2.290	2.373	2.475	2.547	
	Ebn at Noon	283	292	289	279	263	255	241	246	263	265	272	278	
	Edh at Noon	27	30	36	43	49	50	56	51	39	33	27	24	
All-Sky Solar Radiation	RadAvg	864	1068	1415	1775	1938	2030	1929	1749	1499	1238	948	755	
	RadStd	47	114	110	125	156	138	118	82	126	149	77	65	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (71 neighbors)	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+306	+221

Nomenclature: See separate page